

Bayesian Analysis of US Spot Bitcoin ETFs

Market Structure, Return Inference, and Tracking Efficiency

Davide Frosi

Introduction to Bayesian Learning

Academic Year 2025-2026

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1. Introduction

On January 11, 2024, the SEC approved the first US spot Bitcoin ETFs after more than a decade of rejections. Six asset managers launched products the same morning: BlackRock (IBIT), Fidelity (FBTC), Bitwise (BITB), ARK Invest (ARKB), VanEck (HODL), and Invesco (BTCO). They all hold the same underlying asset. They all launched the same day. Their expense ratios differ by at most five basis points.

This near-identical setup is unusual in financial data. Normally when comparing assets, you worry about confounding differences like insect or, currency, leverage and liquidity. Here, all six products are wrappers on spot Bitcoin, traded on US exchanges, priced in dollars. Any systematic differences in their market behaviour must come from fee structure, brand recognition, or institutional distribution. That makes them unusually clean material for statistical inference.

We use 596 trading days of daily prices and volumes (January 2024 to May 2026) to address five questions through a Bayesian lens:

- Has BlackRock's market share advantage changed as the ETF market matured?
- Among the non-IBIT products, is there a genuine leader or is competition roughly even?
- What does the posterior distribution of daily expected returns tell us about market efficiency?
- Can we statistically distinguish the ETFs by how cleanly they track Bitcoin?
- Does IBIT replicate Bitcoin returns faithfully ($\beta = 1$), and has this improved over time?

The analysis is structured around the conjugate families we covered in class: Beta-Binomial for proportion inference, Normal-InvGamma for mean and variance inference, and Bayesian linear regression via MCMC. We use improper Jeffreys priors where conjugate closed-form results exist, and *brms*/Stan for the regression models. All posterior quantities that required simulation used between 5 and 10 million Monte Carlo draws.

2. Data

Daily adjusted closing prices and trading volumes were downloaded directly from Yahoo Finance using the *quantmod* package in R. The sample runs from January 12, 2024 (the first full trading day after approval) through May 29, 2026, giving 596 trading days after removing rows with any missing values — mostly caused by BTCO's thin early trading before it settled into normal operation.

For each ETF we computed:

- Daily log-return: $r_t = \log(P_t / P_{t-1})$
- Daily volume share: $s_{k,t} = \text{volume}_k / (\text{sum of volumes across all 6 ETFs on day } t)$
- Daily tracking error: $TE_{k,t} = r_{k,t}^{\text{ETF}} - r_{k,t}^{\text{BTC}}$

We also split the sample on January 11, 2025, the one-year anniversary of the ETF launch, to study whether the early days of market competition differed from the post-maturity period. Year 1 covers 250 trading days; Year 2+ covers the remaining 346.

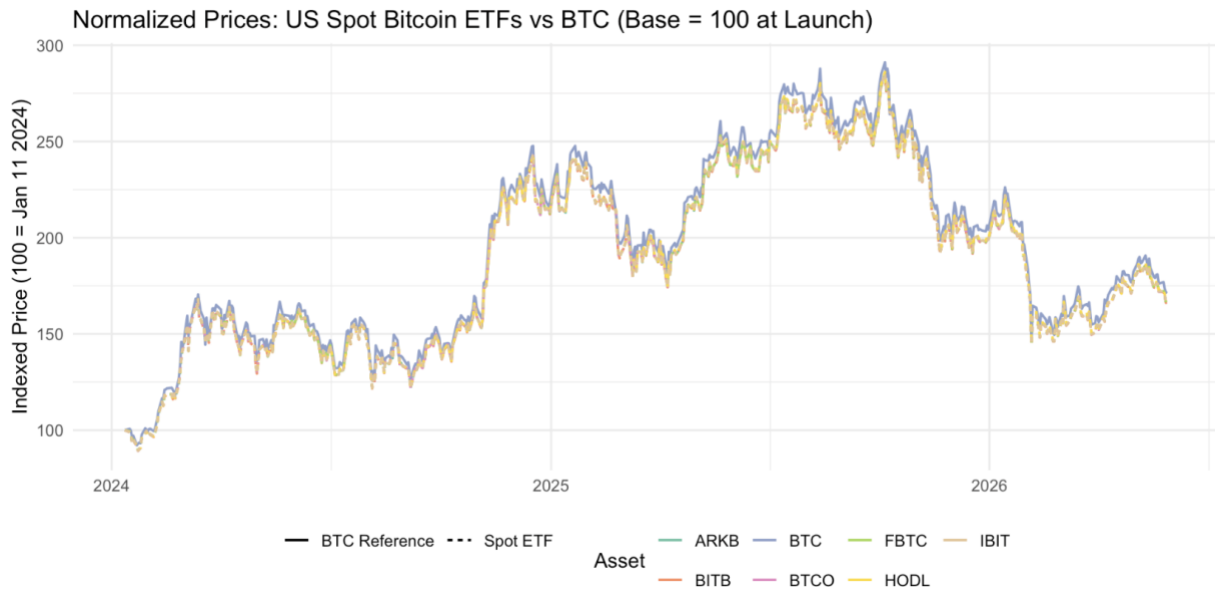
ETF Overview

Ticker	Issuer	Fee (%)	AUM Rank	Year 1 (avg share)	Year 2+ (avg share)
IBIT	BlackRock iShares	0.25	1st	66.5	81.5
FBTC	Fidelity	0.25	2nd	14.2	7.3
BITB	Bitwise	0.20	3rd	4.8	3.2
ARKB	ARK 21Shares	0.21	4th	11.7	5.5
HODL	VanEck	0.20	5th	2.1	2.3
BTCO	Invesco Galaxy	0.25	6th	0.7	0.3

Table 1. The six US spot Bitcoin ETFs studied.

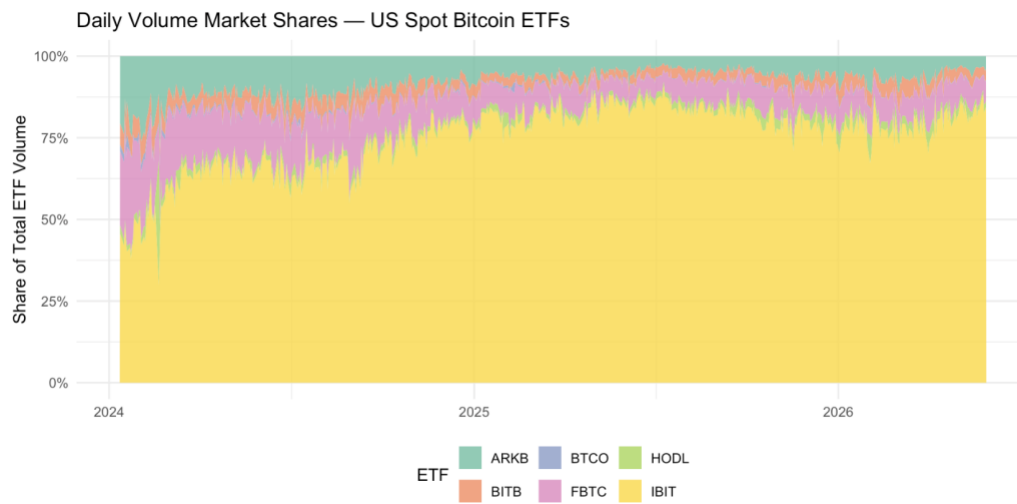
3. Exploratory Analysis

The six ETFs move essentially in lockstep with Bitcoin. Over 596 days the pairwise log-return correlations range from 0.9995 to 0.9998, high enough that the differences are entirely noise at daily frequency. Normalized to a common base of 100 at launch, all six products trace almost the same line as Bitcoin itself, with visible divergence only during the early weeks when volumes were thin and bid-ask spreads were wide.



Normalized price series for all six ETFs and Bitcoin, rebased to 100 at launch (January 12, 2024).

Volume tells a different story. The stacked area chart below shows a market that started fragmented and converged rapidly. In the first weeks, volume was distributed across multiple products. By mid-2024, IBIT and FBTC together accounted for most days, and by Year 2 IBIT alone dominates almost every session.

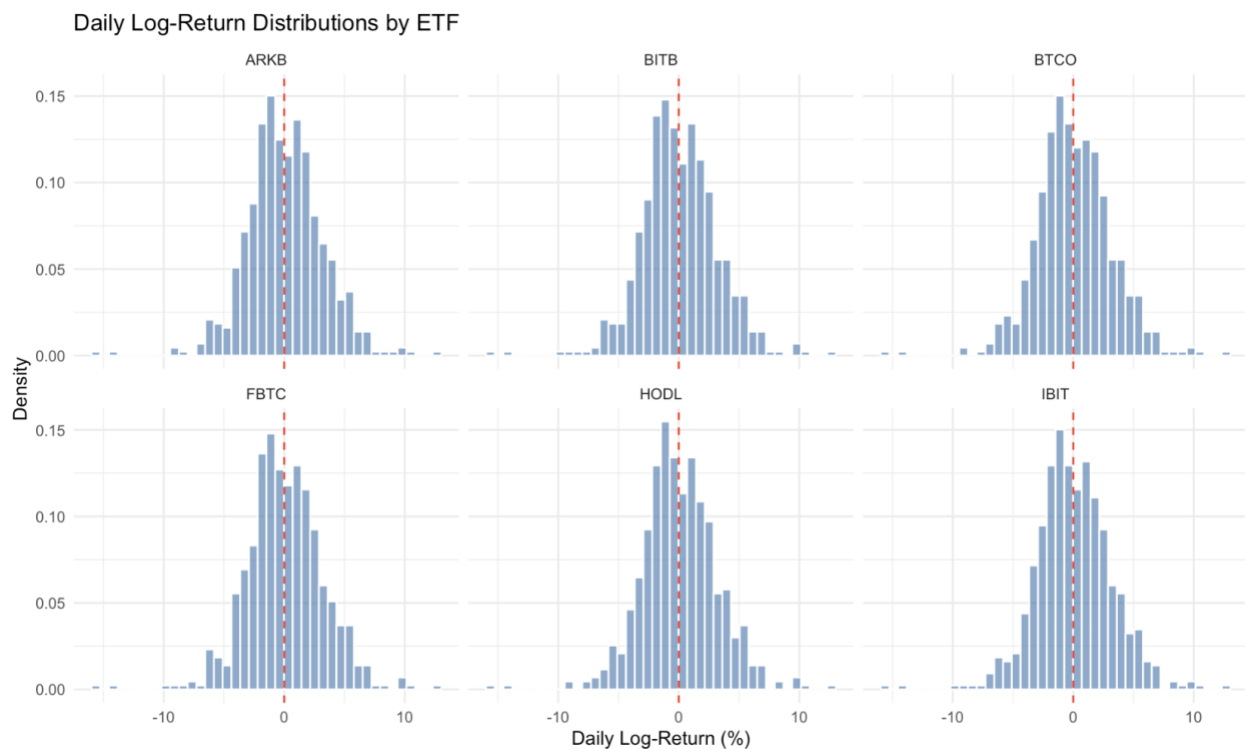


Daily volume market shares for all six ETFs, stacked area format (January 2024 - May 2026).

Summary statistics for daily log-returns confirm what the correlations imply: the six means are virtually identical (all near 0.075% per day, ~19% annualized), as are the standard deviations (~3.1% per day). Skewness and excess kurtosis are mild, consistent with the roughly symmetric but heavy-tailed distribution expected from a high-volatility asset. A normal assumption is good enough for the main analyses.

ETF	N	Mean (%)	SD (%)	Ann. Ret. (%)	Ann. Vol. (%)	Skewness	Ex. Kurt.
IBIT	596	0.0750	3.152	18.89	50.04	-0.065	1.916
FBTC	596	0.0749	3.149	18.89	49.99	-0.072	1.928
BITB	596	0.0747	3.139	18.83	49.83	-0.071	1.975
ARKB	596	0.0749	3.137	18.86	49.79	-0.074	1.925
HODL	596	0.0756	3.133	18.99	49.73	-0.064	1.995
BTCO	596	0.0756	3.126	19.00	49.62	-0.069	1.963

Table 2. Summary statistics for daily log-returns. Means are in percentage points.



4. IBIT Market Dominance and the Regime Shift

IBIT captured more than 50% of total ETF volume on virtually every day in the sample, which makes any proportion test on that threshold degenerate, i.e. the posterior probability rounds to 1. We therefore focus on a sharper question: whether IBIT achieves strong dominance, defined as capturing more than 70% of total volume on a given day.

We model each trading day as a Bernoulli trial where success means IBIT's volume share exceeds 0.70, and we estimate π_1 and π_2 separately for Year 1 and Year 2+. Under a uniform Beta(1, 1) prior, encoding no prior belief about BlackRock's competitive position, the posterior for each period is analytically:

$$\pi_j | y_j \sim \text{Beta}(y_j + 1, n_j - y_j + 1)$$

4.1 Results

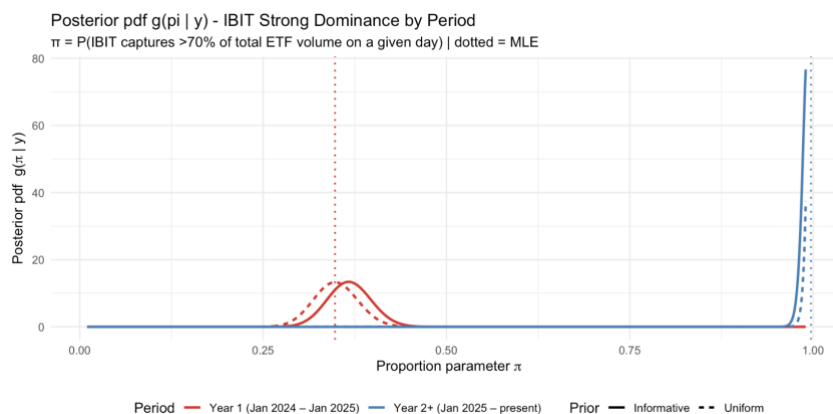
The data surprised us. We had expected that IBIT's initial advantage would erode as competitors caught up and arbitrageurs spread capital more evenly. The opposite happened.

Period	Days (n)	Success (y)	MLE π	Post. Mean	95% HPD
Year 1 (Jan 2024 – Jan 2025)	250	87	0.348	0.349	(0.290, 0.408)
Year 2+ (Jan 2025 – May 2026)	346	345	0.997	0.994	(0.989, 1.000)

Table 3. Posterior inference for IBIT strong dominance (>70% daily volume share). Uniform Beta(1, 1) prior.

In Year 1, IBIT crossed the 70% threshold on only 87 of 250 days, 35% of the time. The 95% HPD interval (0.290, 0.408) is fairly wide, reflecting genuine uncertainty about the underlying rate. In Year 2+, IBIT cleared 70% on 345 of 346 days. The 95% HPD interval is (0.989, 1.000).

The two posterior distributions do not overlap. The difference $\pi_2 - \pi_1$ has a posterior mean of approximately 0.645 with a 95% CI of (0.585, 0.703). The frequentist two-proportion test confirms this: $\chi^2 = 306.6$, p-value $< 2.2 \times 10^{-16}$.



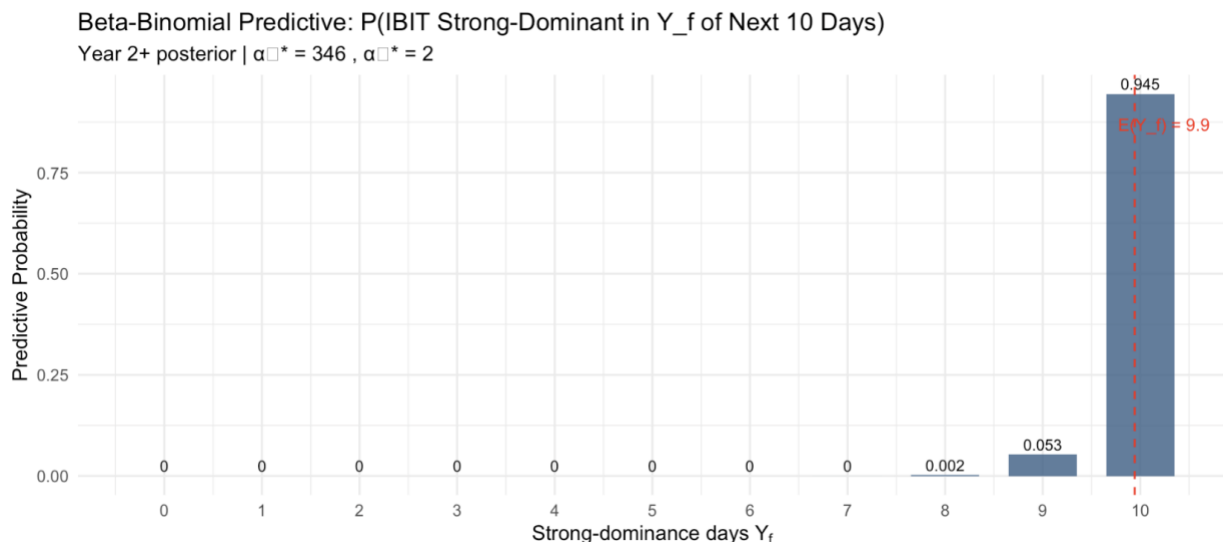
Posterior pdfs for π_1 (Year 1) and π_2 (Year 2+) under uniform and informative priors. The two posteriors are entirely non-overlapping.

4.2 Prior Sensitivity

We fitted the same model under three prior specifications: uniform Beta(1,1), Jeffreys Beta(0.5, 0.5), and an informative Beta(9.5, 3.0) encoding the view that BlackRock's brand advantage would produce strong dominance on 60-90% of days. The informative prior had a prior mean of 0.76, higher than what Year 1 turned out to be. Despite this, all three priors give essentially the same posterior means (0.348-0.368 for Year 1; 0.989-0.994 for Year 2+), with prior weight below 1% in both periods. With $n = 250$ or 346 observations, the data dominates regardless of prior choice.

4.3 Predictive Distribution

Under the Year 2+ posterior, the Beta-Binomial predictive distribution for the next 10 trading days assigns probability 0.945 to all 10 days being strong-dominance days, and essentially zero probability to fewer than 7. The expected number of strong-dominance days in the next 10 is 9.94. In practical terms: IBIT's position in the US spot Bitcoin ETF market has consolidated to the point where a non-dominant day is a statistical anomaly.



Beta-Binomial predictive distribution for the number of IBIT strong-dominance days in the next 10 trading sessions, based on the Year 2+ posterior.

5. Competition in the Second Tier

Given IBIT's near-complete dominance of total volume, the relevant competitive question shifts to the remaining five ETFs: FBTC, BITB, ARKB, HODL, and BTCO. We define a new Bernoulli experiment: on day t , ETF k "wins" if its volume exceeds every other non-IBIT competitor. We estimate $\pi_k = P(\text{ETF } k \text{ is the non-IBIT volume leader on a random trading day})$ for each k , with a fair-share null hypothesis of $\pi = 0.20$.

ETF	Days Led	$\hat{\pi}$	Post. Mean	95% HPD	Excess vs 0.20	$P(\pi > 0.20)$	$P(\pi < 0.20)$
FBTC	473	0.7936	0.7926	(0.760, 0.825)	+0.5926	1.000	0.000
BITB	0	0.0000	0.0017	(0.000, 0.003)	-0.1983	0.000	1.000
ARKB	122	0.2047	0.2057	(0.173, 0.238)	+0.0057	0.628	0.372
HODL	1	0.0017	0.0033	(0.000, 0.007)	-0.1967	0.000	1.000
BTCO	0	0.0000	0.0017	(0.000, 0.003)	-0.1983	0.000	1.000

Table 4. Posterior inference for non-IBIT ETF volume leadership. Uniform Beta(1,1) prior. $P(\pi > 0.20)$ is the posterior probability of exceeding the 1/5 fair-share baseline.

The market has a clear three-tier structure. FBTC dominates the second tier with a posterior mean of 0.7926, leading on roughly four out of every five trading days. At the other extreme, BITB, HODL, and BTCO are effectively non-participants with posterior means near zero.

The one genuinely uncertain result is ARKB. With 122 winning days in 596 trading sessions, its sample proportion is 0.205, barely above the 0.20 fair-share threshold. The posterior probability $P(\pi_{\text{ARKB}} > 0.20) = 0.628$, meaning the data provides moderate but not convincing evidence that ARKB leads more often than chance. Given that ARK Invest has a retail-focused, high-conviction investor base, this slight edge is plausible but hardly established.

6. Bayesian Return Inference

Each ETF's daily log-return series is modelled as an iid normal sample: $Y_i | \mu, \sigma^2 \sim N(\mu, \sigma^2)$. The EMH implies $\mu = 0$, if a positive expected daily return were predictable, arbitrage would eliminate it. We use $\mu_0 = 0$ as our prior mean with $n_0 = 2$ imaginary prior observations, giving the Normal-InvGamma conjugate structure. For σ^2 we use the weakly informative InvGamma(0.5, 0.5) prior, which assigns 95% probability to daily σ between 44.61% and 3,191%, bounds so extreme that the prior is essentially non-informative over any realistic range for daily ETF returns.

6.1 Analytical Posterior via Conjugacy

With conjugate priors, the marginal posterior for μ follows a scaled t-distribution centered at the posterior mean:

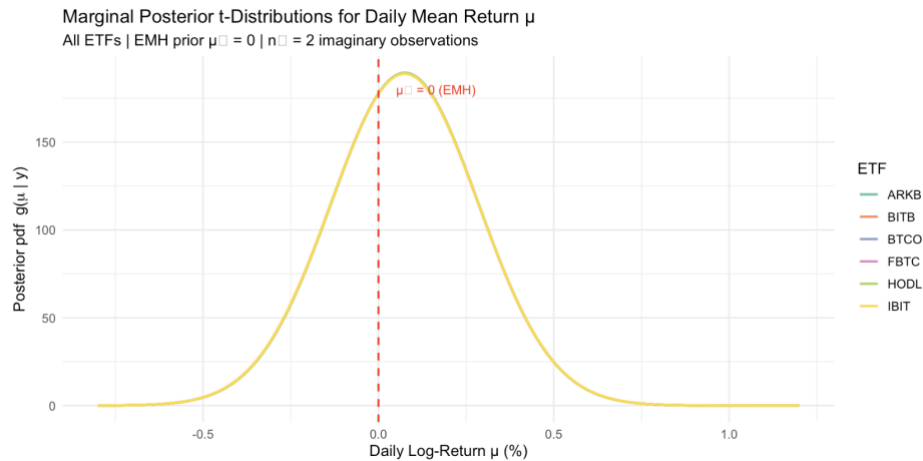
$$E(\mu | y) = [n / (n_0 + n)] \times \bar{y} + [n_0 / (n_0 + n)] \times \mu_0$$

With $n = 596$ and $n_0 = 2$, the prior receives a weight of only 0.33%, so posterior means are essentially identical to sample means.

ETF	Sample Mean	Post. Mean	95% CI Lower	95% CI Upper	$P(\mu > 0)$	Ann. Return (%)
IBIT	0.0007496	0.0007471	-0.003399	0.004893	0.638	18.83
FBTC	0.0007495	0.0007470	-0.003398	0.004892	0.638	18.82
BITB	0.0007473	0.0007448	-0.003395	0.004885	0.638	18.77
ARKB	0.0007485	0.0007460	-0.003393	0.004885	0.638	18.80
HODL	0.0007559	0.0007534	-0.003383	0.004890	0.640	18.99
BTCO	0.0007563	0.0007538	-0.003380	0.004887	0.640	19.00

Table 5. Posterior return inference for all six ETFs. Normal-InvGamma conjugate model with EMH prior $\mu_0 = 0$.

All six ETFs give $P(\mu > 0) \approx 0.638$ -0.640, and all 95% credible intervals contain zero by a wide margin. An annualized return of ~19% sounds large, but daily log-returns are extremely noisy: the standard deviation (~3.4% per day) is roughly 45 times the mean, so 596 observations are simply not enough to reject $\mu = 0$ with any conviction.



Marginal posterior t -distributions for daily mean return μ across all six ETFs. The red dashed line marks $\mu_0 = 0$ (EMH).

6.2 Improper Priors and Frequentist Equivalence

Under an improper Jeffreys prior $p(\mu, \sigma^2) \propto 1/\sigma^2$, the posterior for μ becomes analytically identical to the frequentist t -distribution. The Bayesian 95% credible interval for IBIT's mean daily return is $(-0.00179, +0.00329)$, matching the frequentist confidence interval to five decimal places. The frequentist p -value of 0.562 ($t = 0.58$, $df = 595$) corresponds directly to the posterior tail probabilities $P(\mu < 0) = 0.281$ and $P(\mu > 0) = 0.719$.

6.3 Posterior Predictive Intervals

The 95% posterior predictive interval for IBIT's next single trading day is $(-6.12\%, +6.27\%)$, the full uncertainty about tomorrow's return. For the sample mean of the next 30 trading days, it narrows to $(-1.08\%, +1.23\%)$. This compression is exactly what we could have predicted: as n grows, the prediction interval's half-width converges toward $s/\sqrt{n}$ rather than $s \cdot \sqrt{(1 + 1/n)}$. Intraday Bitcoin can easily move $\pm 6\%$ in a session; the 30-day interval, while narrower, still easily contains zero.

7. Tracking Error Variance

While mean returns are statistically indistinguishable, the variance of the tracking error, $TE_{k,t} = r_{k,t}^{ETF} - r_{t}^{BTC}$, is theoretically more informative. A more liquid ETF with a tighter bid-ask spread and faster creation/redemption mechanism should have lower tracking error variance. We fitted a scaled inverse chi-squared posterior for σ_{TE}^2 per ETF under an improper Jeffreys prior.

7.1 Results

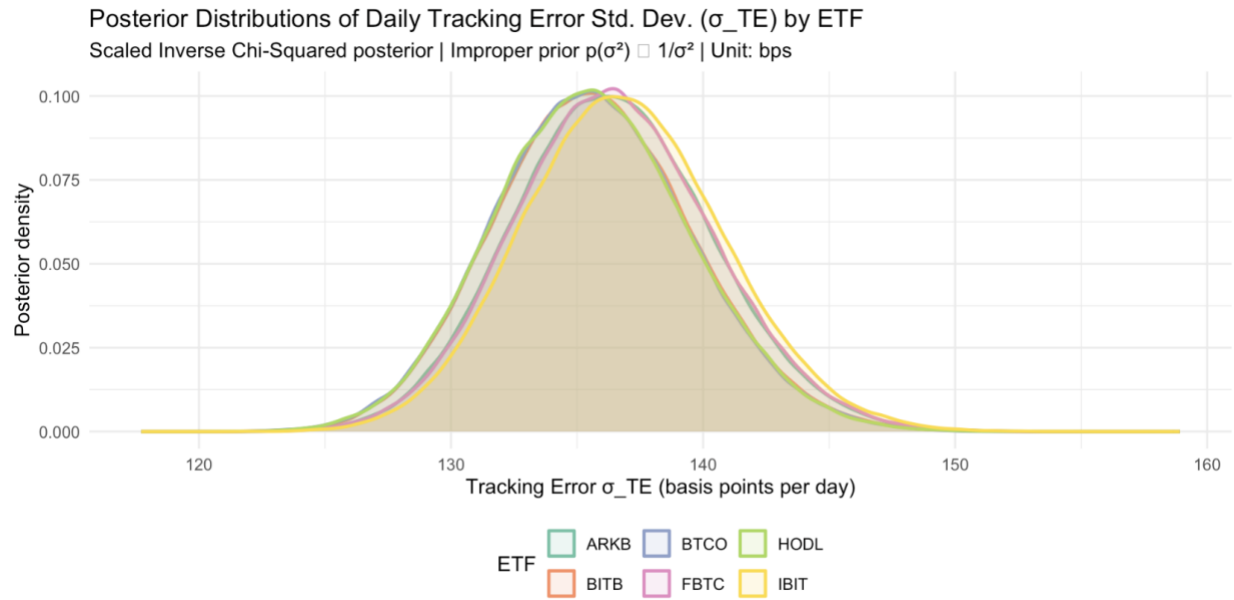
ETF	Fee (%)	Mean TE (bps)	σ_{TE} post. (bps)	95% CI (bps)	Exp. Drag (bps)	Excess Drag (bps)
IBIT	0.25	-0.204	136.93	(129.4, 144.9)	-0.099	-0.105
FBTC	0.25	-0.206	136.54	(129.0, 144.5)	-0.099	-0.106
BITB	0.20	-0.227	135.76	(128.2, 143.7)	-0.079	-0.148
ARKB	0.21	-0.215	136.47	(128.9, 144.4)	-0.083	-0.132
HODL	0.20	-0.141	135.67	(128.2, 143.6)	-0.079	-0.061
BTCO	0.25	-0.137	135.66	(128.2, 143.6)	-0.099	-0.038

Table 6. Tracking error summary. Mean TE and σ_{TE} are in basis points per day (1 bps = 0.01%). Expected daily drag = -Annual Fee / 252. Excess drag = observed mean TE minus expected drag.

All six posterior σ_{TE} estimates cluster tightly around 136 basis points per day. Bartlett's test of variance homogeneity gives $K^2 = 0.096$ with $p = 0.9999$, the test has essentially no power to reject equality. Every pairwise $P(\sigma_{ETF1}^2 < \sigma_{ETF2}^2)$ is between 0.41 and 0.55, all credible intervals for σ ratios contain 1, and all 15 Bonferroni comparisons return "non-significant".

This is a structural limitation of daily closing-price data. The dominant driver of daily tracking error is the overnight Bitcoin move: BTC trades 24 hours a day while ETFs trade only from 9:30am to 4:00pm. An overnight BTC jump of 2% hits all six ETFs identically when they open the next morning. This common-shock component, roughly 136 bps/day in standard deviation, completely swamps the fee-related differences between ETFs, which amount to at most 0.05 bps/day. Distinguishing ETF tracking quality from daily prices would require intraday data or at minimum a model that separates the overnight BTC component.

The more informative metric in this analysis is the mean tracking error rather than the variance. Every ETF exhibits an operational loss that exceeds what its stated fee predicts. For instance, BITB's excess drag is -0.148 bps per day, nearly double the -0.079 bps daily drag implied by its 0.20% expense ratio. Conversely, BTCO and HODL demonstrate remarkably low excess drags of -0.038 bps and -0.061 bps, respectively. While these variations are small in absolute terms, they do not point to a simple liquidity penalty for smaller funds; rather, these differences are likely driven by historical fee waivers or structural nuances in how these specific issuers manage daily creation / redemption mechanics.



Posterior distributions of daily tracking error standard deviation σ_{TE} per ETF. The distributions overlap completely, confirming no statistical distinction is possible at daily frequency.

8. Tracking Regression: Does IBIT Faithfully Follow Bitcoin?

We model IBIT's daily log-return as a linear function of Bitcoin's log-return:

$$\text{IBIT}_{\text{return}} = \alpha + \beta \cdot \text{BTC}_{\text{return}} + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

Under perfect replication, $\alpha = 0$ and $\beta = 1$. The prior on β was $N(1, 0.1)$, encoding that we expect near-unity tracking but allow meaningful uncertainty. The intercept α received $N(0, 0.001)$, consistent with a small daily fee drag and no expectation of alpha generation.

8.1 Full-Sample Results

The posterior estimate of β is 0.932 with a 95% credible interval of (0.897, 0.966). This is far below 1. The posterior probabilities make the finding concrete:

- $P(\beta < 1.00 \mid \text{data}) = 0.9999 \rightarrow$ near-certain systematic under-tracking
- $P(\beta < 0.95 \mid \text{data}) = 0.846 \rightarrow$ strong evidence the gap exceeds 5%
- $P(\beta < 0.93 \mid \text{data}) = 0.452 \rightarrow$ uncertain whether it goes below 0.93
- $P(\alpha > 0 \mid \text{data}) = 0.394 \rightarrow$ no evidence of positive alpha; consistent with EMH

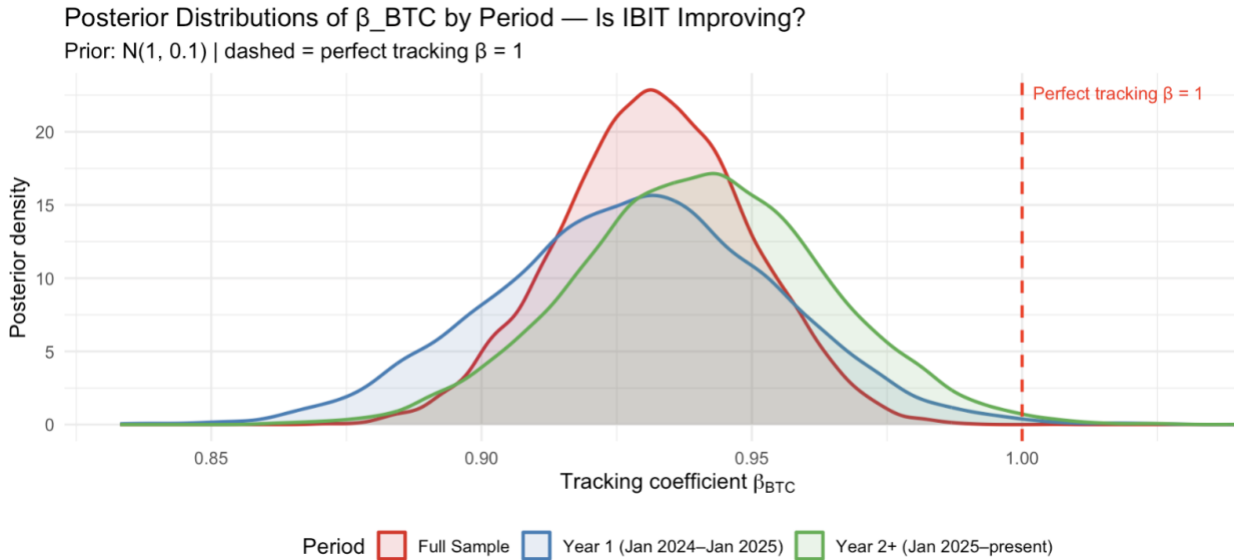
The R^2 of 0.817 is the other surprising number. A spot ETF holding nothing but Bitcoin should explain virtually all of Bitcoin's daily price variance: we would expect R^2 above 0.99. Getting 0.817 means roughly 18% of IBIT's daily return variance is unrelated to Bitcoin's move that day. The explanation is structural: BTC trades continuously, while IBIT trades during US market hours only. A large Bitcoin move between Friday's 4pm close and Monday's 9:30am open appears as a gap in IBIT's opening price, correctly captured the next day, but creating a lag that depresses the same-day β and R^2 .

8.2 Time-Split: Has Tracking Improved?

Period	β (post. mean)	95% CI	$P(\beta < 1.00)$	$P(\beta < 0.95)$
Year 1 (Jan 2024 – Jan 2025)	0.928	(0.878, 0.979)	0.997	0.796
Year 2+ (Jan 2025 – May 2026)	0.940	(0.894, 0.986)	0.994	0.659
Full sample	0.932	(0.897, 0.966)	0.9999	0.846

Table 7. Posterior β_{BTC} estimates by period. Prior: $N(1, 0.1)$ for β , $N(0, 0.001)$ for intercept.

β nudged upward from 0.928 in Year 1 to 0.940 in Year 2+, and $P(\beta_{Y2} > \beta_{Y1} \mid \text{data}) = 0.641$, moderate directional evidence of improvement but not conclusive. This is consistent with arbitrage becoming more efficient as IBIT's AUM grew, making it easier for authorized participants to keep the ETF price aligned with its NAV (Net Asset Value) throughout the day. The gap from 1 remains substantial in both periods.



Posterior distributions of β_{BTC} for Year 1, Year 2+, and full sample. The vertical dashed line marks perfect tracking $\beta = 1$. None of the three posteriors cover 1 with any meaningful probability.

8.3 Volume Share as an Additional Predictor

We added IBIT's daily volume share as a second predictor to test whether days with unusually high IBIT trading volume correspond to a price premium. The posterior coefficient on volume share is essentially zero: mean 0.000, 95% CI $(-0.007, +0.014)$, $P(\gamma > 0) = 0.758$. CI contains zero; volume share adds no explanatory power, it does not predict IBIT returns after conditioning on Bitcoin.

8.4 Prior Sensitivity

Replacing the informative $N(1, 0.1)$ prior on β with a diffuse $N(0, 10)$ changes the posterior mean by 0.002 (0.932 vs 0.929) and barely shifts the credible interval. With 596 daily observations, the data completely overwhelms any reasonable prior on β .

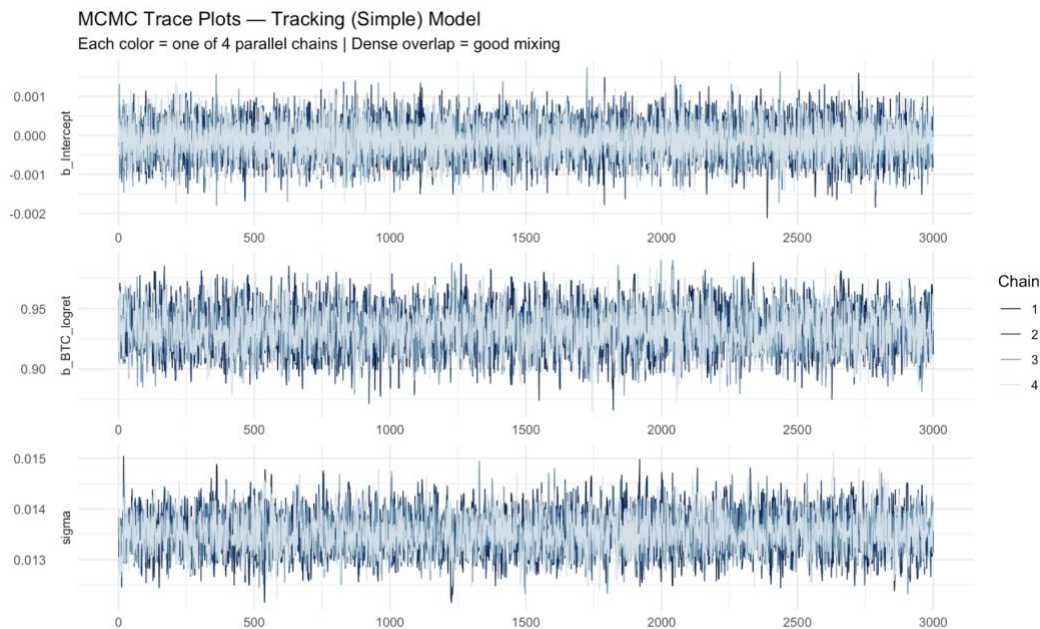
9. MCMC Diagnostics

All four brms models (simple tracking, extended tracking, time-split Year 1, time-split Year 2+) were fitted with 4 parallel chains of 4,000 iterations each (1,000 warmup). Convergence was assessed via R-hat and effective sample size (ESS).

Model	Max R-hat	Any R-hat > 1.01?	Min Bulk ESS	Status
Tracking (simple)	1.0030	No	5,480	✓ OK
Tracking (extended)	1.0008	No	7,186	✓ OK
Time-split Year 1	1.0003	No	5,293	✓ OK
Time-split Year 2+	1.0006	No	5,188	✓ OK

Table 8. MCMC convergence diagnostics. $R\text{-hat} \leq 1.01$ and Bulk ESS > 400 are the standard thresholds for reliable inference.

All four models pass both criteria easily. R-hat never exceeds 1.003, and the minimum Bulk ESS across all models is 5,188, more than ten times the threshold of 400 required for reliable central estimates. The trace plots show dense, stationary caterpillar patterns with no drift or between-chain separation. The autocorrelations plots show rapid decay for all parameters, with negligible correlation evident by approximately lag 5, consistent with efficient Hamiltonian Monte Carlo (HMC) sampling.



MCMC trace plots for the simple tracking model (intercept, β_{BTC} , σ). Four chains, 3,000 post-warmup iterations each.

10. Discussion and Conclusions

This project used six US spot Bitcoin ETFs as a Bayesian laboratory. Because all products hold the same asset, launched simultaneously, and trade in the same market, differences in their behaviour isolate brand, fee, and liquidity effects with unusual clarity.

What we found

On market structure, IBIT's dominance intensified rather than eroded. In Year 1, it cleared the 70% strong-dominance threshold on only 35% of trading days. By Year 2+, it cleared it on 99.7% of days. The two posterior Beta distributions don't overlap at all. Whatever competitive threat the other products posed in early 2024 had effectively collapsed by early 2025. Fidelity retained a clear second-tier position, ARK sits near fair share, and the remaining three products are statistical noise.

On return inference, the EMH prior $\mu_0 = 0$ receives very little updating from 596 days of Bitcoin returns. $P(\mu > 0)$ sits at 0.64-0.72 depending on the prior, evidence of a positive expected return, but far from conclusive. The wide predictive intervals ($\pm 6\%$ for a single day) put this in context. The numerical identity between Bayesian posterior intervals under improper priors and frequentist confidence intervals, demonstrated cleanly in Section 6.2, is one of the more pedagogically satisfying results, as we have seen it in class.

On tracking quality, we were unable to rank ETFs by tracking error variance from daily data. The posterior distributions of σ_{TE} for all six products are essentially identical, because the overnight Bitcoin move that hits every ETF at market open each morning, roughly 136 basis points per day in standard deviation, drowns out the fee-related differences between them. This is not a modelling failure; it is what the data contain. What the mean tracking error does reveal is that some ETFs (notably BITB) absorb hidden execution costs above their stated fee, while others (HODL, BTCO) come closer to their stated drag.

On the regression, the most concrete finding in the analysis is $\beta = 0.932$ with $P(\beta < 1) = 0.9999$. A spot ETF that holds Bitcoin should have $\beta = 1$ and R^2 above 0.99. It has neither. The 18% of IBIT's daily return variance unexplained by Bitcoin's same-day return is a direct consequence of the 24/7 BTC market versus the 6.5-hour ETF trading window: large overnight Bitcoin moves show up in the following day's IBIT open, creating a structural lag that inflates tracking error and depresses β . There is no evidence of alpha ($P(\alpha > 0) = 0.39$), consistent with the view that a passive wrapper on spot Bitcoin cannot systematically outperform its benchmark.

Limitations

Three limitations are worth noting. First, daily closing prices cannot resolve intraday arbitrage quality; a study using bid-ask spreads and intraday quote data would give a much richer picture of tracking efficiency. Second, the sample covers only the first two years of a genuinely new product class. Market dynamics may continue to shift; the regime-shift finding is a snapshot, not a long-run equilibrium. Third, we worked under the simplifying assumption that daily log-returns are iid normal. The heavy tails visible in the histograms suggest that a Student-t likelihood would

be more robust, though it would come at the cost of the clean conjugate structure that made Sections 4–7 tractable.

Appendix: Prior Specifications

Table A1 summarizes all prior distributions used across the analysis.

Parameter	Prior family	Hyperparameters	Justification
π (IBIT dominance), Uniform	Beta(1,1)	$\alpha_1=1, \alpha_2=1$	No prior knowledge; standard reference for proportions
π (IBIT dominance), Jeffreys	Beta(0.5, 0.5)	$\alpha_1=0.5, \alpha_2=0.5$	Objective reference prior derived from Fisher information
π (IBIT dominance), Informative	Beta(9.5, 3.0)	$\alpha_1=9.5, \alpha_2=3.0$	$P(\pi < 0.60) = 0.10, P(\pi < 0.90) = 0.90$; encodes brand advantage prior belief
μ (daily return), conjugate	$N(\mu_0, \sigma^2/n_0)$	$\mu_0=0, n_0=2$	EMH prior; zero expected return; 2 imaginary obs.
σ^2 (daily return)	InvGamma(0.5, 0.5)	$\alpha=0.5, \beta=0.5$	Weakly informative; $P(\sigma \in (0.45\%, 3191\%)) = 0.95$
σ^2_{TE} (tracking error)	Improper Jeffreys	$p(\sigma^2) \propto 1/\sigma^2$	Reference prior; posterior = scaled inverse chi-squared
α (regression intercept)	$N(0, 0.001^2)$	$\mu=0, \sigma=0.001$	Tiny fee drag expected; no alpha for passive ETF
β (regression slope)	$N(1, 0.1^2)$	$\mu=1, \sigma=0.1$	Near-unity tracking expected; allows ± 0.3 range at 3σ
σ (regression noise)	Exponential(100)	rate=100	$P(\sigma < 0.01) \approx 0.63$; tracking error $\sim 1\%$ daily

Table A1. Complete prior specifications for all models in the analysis.